

# Robustness of Bayesian Neural Networks Design Choices Against Weight and Bias Perturbation Induced by Single Event Upsets (SEUs)

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**Abstract:** Bayesian Neural Networks (BNNs) naturally model uncertainty in safety-critical environments, yet their resilience against hardware-induced Single Event Upsets (SEUs) remains under-studied compared to input perturbations. This paper presents the first systematic evaluation of BNN architectural choices under internal weight and bias bit-flips. We evaluate an exhaustive SEU grid across two satellite imaging datasets (ShipsNet, EuroSAT) using 5-fold cross-validation. To capture both failure modes, we measure degradation using the Aggregate Robustness Index (ARIn), combining accuracy drop and confidence collapse. Our findings resolve previous metric ambiguities: deterministic CNNs are strictly more robust to parameter corruption than BNNs, and structural regularizers offer no measurable SEU defense. Instead, the activation function dictates resilience. The bounded Weighted Gaussian activation simultaneously improves EuroSAT robustness by 46% (ARIn 0.0713 vs. 0.1317 for ReLU) and accuracy (+7%). Crucially, we isolate the IEEE-754 exponent MSB (bit 1) as the catastrophic failure mode across all architectures (ARIn 0.332 vs.  $\approx 0.05$  for mantissa bits). These results prove that targeted exponent-hardening and bounded activations provide massive resilience against SEUs, offering a highly efficient alternative to full-word error correction for spaceborne AI.

## 1 Introduction

Machine learning models are increasingly deployed in harsh environments with unique reliability demands. A representative example is CloudScout, a CNN-based system onboard the ESA PhiSat-1 mission that performs on-board cloud detection to reduce transmission bandwidth. In space and other high-radiation settings, these models are exposed to Single Event Effects (SEEs) caused by ionizing radiation. A critical manifestation is a Single Event Upset (SEU), a transient bit-flip in memory where model parameters are stored. These bit-flips pose a severe reliability threat, capable of instantly degrading the performance of safety-critical ML systems.

Prior work has extensively studied the robustness of both deterministic and Bayesian Neural Networks (BNNs) against input-perturbation, such as adversarial attacks. However, robustness against internal weight and bias perturbations induced by hardware-level SEUs remains under-explored. While recent re-

search has begun addressing model-intrinsic robustness for deterministic CNNs under weight perturbation, there has been little to no investigation into how the unique probabilistic nature of BNN parameters responds to such bit-level corruption.

To address these gaps and formally evaluate hardware-level resilience, our primary contributions are:

1. The first systematic study of BNN design choices (priors, regularizers, activations) under internal weight and bias SEUs.
2. The deployment of the Aggregate Robustness Index (ARIn), a composite metric penalizing both accuracy loss and confidence collapse.
3. Rigorous evaluation across two datasets (ShipsNet, EuroSAT) under 5-fold cross-validation with statistical confidence intervals, proving results are not optimization artifacts.
4. The identification of the exponent MSB (bit 1) as the catastrophic failure mode, yielding a concrete targeted-hardening recommendation.

The remainder of this paper is structured as follows. Section 2 formalizes the SEU problem and reviews related work. Section 3 outlines our method-

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ology and the SEU injection protocol. Section 4 presents the comparative robustness results across datasets, and Section 5 concludes with recommendations for safety-critical deployment.

## 2 Background

### 2.1 Single Event Effects and Upsets

Single Event Effects (SEEs) happen when a single ionizing particle penetrates sensitive nodes in an electronic device. This can induce a Single Event Upset (SEU), temporarily flipping a memory or control bit. Model weights in PyTorch are stored in the standard IEEE-754 FP32 format, comprising 1 sign bit, 8 exponent bits, and 23 mantissa bits. Because of this layout, the impact of a bit-flip is highly asymmetric: a flip in the mantissa perturbs the value fractionally, whereas a flip in the most significant bits of the exponent (e.g., bit 1) can multiply the parameter by up to  $2^{128}$ . This structural vulnerability sets the stage for the catastrophic exponent failures identified in Section 4.

### 2.2 Fault Injection in Neural Networks

SEUs are simulated by injecting transient bit-flips into model weights during inference. For deterministic neural networks, Single Event Upset Induced Parameter Perturbation (SIPP) has been shown to drastically degrade accuracy, prompting hardware-level protections like Triple Modular Redundancy (TMR) or Error-Correcting Codes (ECC). A more recent state-of-the-art approach to model-intrinsic robustness proposes novel bounded activation functions (like Weighted Gaussian) to limit the propagation of flipped bits. While these existing methods focus on deterministic architectures, our work differs by rigorously evaluating whether the probabilistic nature of Bayesian Neural Networks naturally provides similar intrinsic resilience against weight perturbations.

### 2.3 Bayesian Neural Networks

Unlike deterministic networks that learn point estimates for weights, Bayesian Neural Networks (BNNs) learn a posterior probability distribution  $p(\theta | \mathcal{D})$  over the weights  $\theta$  given the training data  $\mathcal{D}$ . Due to the intractability of the true posterior, this is typically approximated using Variational Inference (VI), optimizing a surrogate distribution by maximizing the Evidence Lower Bound (ELBO). During inference,

predictions are made via Monte Carlo (MC) sampling, drawing multiple weight configurations from the posterior and averaging their outputs, allowing the model to express uncertainty.

## 2.4 Problem Formulation

Formally, let  $f_\theta$  denote a neural network parameterized by  $\theta$ . An SEU is a mapping  $\theta \mapsto \theta'$  that flips a specific bit  $i$  of a single scalar parameter at site  $s$ . We define the robustness of  $f_\theta$  as the expected degradation of a predictive functional (such as accuracy or confidence) over a defined distribution of injection sites  $s$  and bit indices  $i$ . A more robust architecture minimizes this expected degradation when subjected to the  $\theta \mapsto \theta'$  mapping.

## 3 Methodology

### 3.1 Datasets

To ensure findings generalize across tasks, we evaluate our models on two distinct satellite image datasets. The primary dataset is ShipsNet (binary classification of ship vs. no-ship), containing 4,000 RGB images resized to  $64 \times 64$  pixels. The secondary dataset is EuroSAT, comprising 27,000  $64 \times 64$  multispectral images spanning 10 land-cover classes. Both datasets are standardized. For ShipsNet, we use 5-fold cross-validation based on an initial 80/20 train/test split. For EuroSAT, training incorporates likelihood scaling (`obs_scale=400`) to handle the 10-class complexity, meaning the two datasets optimize mathematically distinct ELBO formulations.

### 3.2 Model Architectures and Design Choices

The baseline architecture is a CNN comprising two convolutional layers (`conv1`:  $3 \rightarrow 32$ ,  $k=3$ , and `conv2`:  $32 \rightarrow 64$ ,  $k=3$ ), each followed by pooling, and a fully connected layer (`fc1`). The deterministic baseline and BNN share this structure. We evaluate four architectural variants: Base, Smart Pooling (discarding activations above a threshold), Dropout ( $p = 0.5$ ), and Weight Decay ( $\lambda = 10^{-4}$ ). The BNN expands this grid across seven activation functions, three prior distributions (Gaussian, Laplace, Uniform), and three prior scales ( $b \in \{10.0, 1.0, 0.1\}$ ).

### 3.3 SEU Injection Protocol

SEUs are injected by flipping a single bit in the IEEE-754 FP32 representation of a model parameter during inference. The injection grid spans across target layers (`conv1`, `conv2`, `fc1`), parameter locations (first and last neurons), parameter types (weight, bias), and selected bit indices (0, 1, 3, 6, 10, 15, 21). The exhaustive grid across all configurations yields 42,336 injections for EuroSAT and 14,112 injections per fold for ShipsNet. However, we explicitly exclude degenerate injections that flip the sign bit of a prior scale/variance parameter, as negative variance is invalid. This exclusion removes 3,024 injections from EuroSAT and 1,008 from ShipsNet, ensuring a clean and mathematically valid evaluation.

### 3.4 Robustness Metrics

We measure robustness primarily through two metrics: Absolute Accuracy Difference (AAD) and Average Softmax Difference (ASD). AAD captures the hard drop in classification accuracy, while ASD measures the shift in prediction confidence.

$$\text{AAD} = |a_{\text{after}} - a_{\text{before}}| \quad (1)$$

$$\text{ARIn} = \sqrt{\frac{\text{AAD}^2 + \text{ASD}^2}{2}} \quad (2)$$

To provide a single unified metric for ranking architectures, we introduce the Aggregate Robustness Index (ARIn). A combined index is necessary because AAD and ASD measure distinct failure modes; a model might maintain its top-1 accuracy while its confidence completely collapses, which is highly dangerous in safety-critical applications. ARIn is computed as the quadratic mean of AAD and ASD, a form deliberately chosen to penalize a large degradation in either component without allowing a small degradation in the other to average it away. However, we acknowledge an empirical limitation: because ASD operates on a wider continuous scale, it often exhibits larger magnitudes than AAD, causing ARIn to be heavily dominated by confidence collapse. To ensure transparency, we report AAD and ASD alongside ARIn throughout Section 4, and refrain from using ARIn to adjudicate cases where the two underlying metrics disagree.

### 3.5 Cross-Validation and Statistical Protocol

To establish that the reported effects are not artefacts of a single random seed or optimisation path, every model is trained under stratified  $k$ -fold cross-validation. We use  $k = 5$  rather than the  $k = 10$  suggested in review. The reason is that a fold in this study is substantially more expensive than a fold in a conventional classification experiment: each additional fold requires not only retraining the full design grid, but also re-running the complete SEU injection sweep against every resulting model, with Monte Carlo inference at each injection site. Measured on our hardware, one configuration takes a median of 6 minutes to train for 100 epochs and a further median of 5 minutes for its 168-injection sweep at  $S = 10$  Monte Carlo samples. Across the design grid of  $7 \times 3 \times 3 \times 4 = 252$  configurations,  $k = 5$  therefore amounts to roughly ten GPU-days;  $k = 10$  would double that without changing the qualitative conclusions, since the fold-to-fold spread we observe is already small relative to the effects being tested. We regard  $k = 5$  as the standard defensible choice under this constraint, and report the resulting interval widths openly so the reader can judge the precision obtained.

Fold 1 is the original 80/20 stratified split used in the submitted version. We verified that this split is a valid member of a  $k = 5$  stratified partition and pinned it as fold 1, so that the previously trained models remain exactly reusable rather than being discarded and regenerated. Folds 2–5 are disjoint and cover the remainder of the dataset.

The unit of analysis is the fold: one fold contributes one observation, so  $n = 5$ . Each fold is first reduced to a single mean over its fixed injection grid, and the interval is then taken over those  $k$  fold means,

$$\text{CI}_{95\%} = \bar{x} \pm t_{0.975,4} \frac{s}{\sqrt{n}}, \quad t_{0.975,4} = 2.776 \quad (3)$$

Crucially, confidence intervals are computed over the  $n = 5$  cross-validation folds, not over the individual SEU injection sites. The injection sites constitute an exhaustive deterministic grid rather than a random sample; taking an interval over them would incorrectly measure the model’s site-to-site heterogeneity rather than its training-run variability. Furthermore, when evaluating two-condition claims (such as comparing BNN versus deterministic CNN baselines, or Weighted Gaussian versus ReLU activations), we analyze the paired per-fold difference. Because all models are trained and evaluated on the exact same five data splits, paired testing ensures that the shared fold-difficulty effect cancels out, yielding a statistically

rigorous comparison.

## 4 Results

### 4.1 Overall Robustness

Aggregated over the full injection grid, the ShipsNet Bayesian models attain a pre-injection accuracy of  $0.9053 \pm 0.0036$  and an ARIn of  $0.0736 \pm 0.0028$ , decomposing into AAD  $0.0321 \pm 0.0013$  and softmax difference  $0.0965 \pm 0.0040$ . Because  $n = 5$  is small enough that an interval conveys little more than the observations themselves, we report the individual fold means directly: ARIn is 0.0768, 0.0738, 0.0718, 0.0746 and 0.0711 for folds 1–5 respectively. The spread across independent training runs is thus roughly 0.006, an order of magnitude smaller than the design-choice and injection-site effects reported below, which is the evidence that those effects are not artefacts of a single seed or optimisation path. Table 1 reports the same decomposition by design choice for both datasets.

[ $\approx 2$  lines] Add the matching EuroSAT sentence once its baseline accuracy is re-evaluated from the best checkpoints. Every figure and table in this section MUST be cited in the text — R1-8 and R1-9 flagged Figures 3, 4 and Table 1 as uncited. Grep the sources for `ref{fig:}` and `ref{tab:}` before submitting.

### 4.2 Effect of Bit Position

The position of the Single Event Upset injection dictates the severity of the fault more than any hyperparameter or architectural choice. On both EuroSAT and ShipsNet, bit 1 (the MSB of the exponent) causes catastrophic failure, yielding an ARIn of 0.332 for EuroSAT and 0.306 for ShipsNet. This impact is roughly an order of magnitude worse than flips in any mantissa bit (which sit at an ARIn of  $\approx 0.05$  for EuroSAT and  $\approx 0.03$  for ShipsNet). The mechanism is structural to the IEEE-754 format: flipping the exponent MSB rescales the parameter by a massive factor of  $2^{128}$ , whereas mantissa flips only perturb the value fractionally.

### 4.3 Effect of Activation Function

In EuroSAT, the Weighted Gaussian (WG) activation function attains both the lowest ARIn (0.0713 vs. 0.1317 for ReLU, an improvement of 46%) and the highest initial accuracy (0.8125 vs. 0.7423). However, this dominance does not completely replicate on ShipsNet. As shown in Table 1, the Sinusoidal

Table 1: Robustness by design choice across both datasets. ShipsNet values are means over 5-fold cross-validation with 95% confidence intervals (Student- $t$ , 4 d.f.); each fold contributes one observation, obtained by averaging over all injection sites for that fold, so intervals reflect variability across independent training runs rather than across injection sites. The ShipsNet columns cover the base variant at  $b=1.0$ ; entries marked n/e were not evaluated under cross-validation, so the prior-scale comparison rests on EuroSAT alone. Accuracy is measured before injection and is therefore undefined for the layer rows. Lower ARIn is more robust.

| Factor         | Level                 | ShipsNet (5 folds)       |              | EuroSAT      |              |
|----------------|-----------------------|--------------------------|--------------|--------------|--------------|
|                |                       | ARIn [95% CI]            | Acc.         | ARIn         | Acc.         |
| Activation     | WG                    | .0600 $\pm$ .0030        | .9242        | <b>.0665</b> | <b>.8482</b> |
|                | RWG                   | .0779 $\pm$ .0021        | .8923        | .0710        | .8217        |
|                | sin                   | <b>.0572</b> $\pm$ .0047 | <b>.9483</b> | .0754        | .8250        |
|                | tanh                  | .0804 $\pm$ .0144        | .8501        | .0840        | .8181        |
|                | sigmoid               | .0692 $\pm$ .0104        | .8602        | .1159        | .6777        |
|                | relu6                 | .0656 $\pm$ .0013        | .9224        | .1194        | .7829        |
|                | relu                  | .1049 $\pm$ .0031        | .9396        | .1298        | .7811        |
| Variant        | Base                  | .0720 $\pm$ .0028        | .9053        | .0948        | .7953        |
|                | Dropout               | <b>.0683</b> $\pm$ .0008 | <b>.9092</b> | <b>.0929</b> | .7749        |
|                | Smartpool             | .0686 $\pm$ .0010        | .9059        | .0955        | <b>.8033</b> |
|                | Weight Dcy.           | .0722 $\pm$ .0061        | .9038        | .0944        | .8006        |
| Prior          | Gaussian              | .0722 $\pm$ .0006        | <b>.9328</b> | .0975        | <b>.7961</b> |
|                | Laplace               | .0832 $\pm$ .0027        | .9188        | .1038        | .7913        |
|                | Uniform               | <b>.0654</b> $\pm$ .0072 | .8643        | <b>.0822</b> | .7933        |
| Injection Site | conv1.weight          | .0634 $\pm$ .0031        | —            | .0887        | —            |
|                | conv1.bias            | <b>.0416</b> $\pm$ .0035 | —            | .0810        | —            |
|                | conv2.weight          | .0524 $\pm$ .0030        | —            | <b>.0719</b> | —            |
|                | conv2.bias            | .0441 $\pm$ .0039        | —            | .0799        | —            |
|                | fc1.weight            | .1136 $\pm$ .0021        | —            | .1056        | —            |
|                | fc1.bias              | .1265 $\pm$ .0034        | —            | .1404        | —            |
| Bit Position   | 0                     | .0313 $\pm$ .0038        | —            | .0615        | —            |
|                | 1                     | .3104 $\pm$ .0043        | —            | .3293        | —            |
|                | 3                     | .0304 $\pm$ .0033        | —            | .0544        | —            |
|                | 6                     | .0306 $\pm$ .0035        | —            | .0617        | —            |
|                | 10                    | <b>.0303</b> $\pm$ .0032 | —            | .0511        | —            |
|                | 15                    | .0304 $\pm$ .0033        | —            | .0511        | —            |
| Original Bit   | 0 (0 $\rightarrow$ 1) | .1223 $\pm$ .0025        | —            | .1338        | —            |
|                | 1 (1 $\rightarrow$ 0) | <b>.0303</b> $\pm$ .0033 | —            | <b>.0532</b> | —            |

(sin) activation strictly dominates WG on ShipsNet in both robustness (ARIn 0.0572 vs. 0.0600) and accuracy (94.83% vs. 92.42%). While WG still provides a significant robustness advantage over standard unbounded activations like ReLU (ARIn 0.1049), the optimal resilient activation appears to be dataset-dependent. Crucially, periodic and bounded activations (Sinusoidal, WG) consistently outclass unbounded ReLUs across both datasets.

### 4.4 Bayesian versus Deterministic Models

Contrary to the hypothesis that Bayesian Neural Networks offer intrinsic resilience through weight uncertainty, our corrected evaluations demonstrate that deterministic CNNs are strictly more robust against internal parameter corruption.

This finding resolves an apparent metric disagree-

ment raised during peer review, where BNNs previously appeared to suffer worse accuracy drops (AAD) but degrade more gracefully in confidence (ASD). Rigorous diagnostic checks revealed this discrepancy was an artefact of two factors: an implementation defect that artificially penalized the deterministic model’s softmax outputs, and an inherent Monte-Carlo sampling noise floor in the BNN that accounts for roughly half of its reported ASD signal.

Once the deterministic baseline was corrected, it demonstrated strict dominance. The deterministic CNN not only maintains higher initial accuracy (97.55% vs.  $\approx 93\%$ ), but it suffers roughly half the absolute accuracy drop (AAD 0.0145 vs.  $\sim 0.03$ ) and significantly less confidence destabilization (ASD 0.0167 vs.  $\sim 0.08$ ) than its Bayesian counterparts. Its composite ARIn of 0.0156 drastically outperforms even the most robust BNN variant (Sinusoidal, ARIn 0.0572).

Table 2: Direct comparison of Deterministic and Bayesian CNN baselines on ShipsNet. The deterministic CNN is strictly superior across all robustness metrics and initial accuracy.

| Architecture        | AAD          | ASD          | ARIn         | Acc.         |
|---------------------|--------------|--------------|--------------|--------------|
| Deterministic CNN   | <b>.0145</b> | <b>.0167</b> | <b>.0156</b> | <b>.9755</b> |
| Bayesian CNN (Base) | .0321        | .0965        | .0720        | .9053        |

We conclude that while BNNs are well-documented to resist *external* input perturbations, their complex internal dynamics and reliance on sampled variance parameters make them mathematically more fragile to *internal* bit-level hardware faults.

#### 4.5 Design Choices with No Measurable Effect

Contrary to earlier claims, structural regularizations provide no measurable robustness benefit against SEUs, constituting a null result for model variants (ablation study). The four model variants span an ARIn of 0.0969–0.0983 on EuroSAT and 0.0698–0.0739 on ShipsNet, representing relative spreads of less than 5%.<sup>1</sup> This indicates that techniques traditionally designed to prevent overfitting do not inherently protect against bit-level parameter corruption.

<sup>1</sup>EuroSAT ARIn: base 0.0978, dropout 0.0969, smart pooling 0.0983, weight decay 0.0980. ShipsNet ARIn: dropout 0.0698, smartpool 0.0701, base 0.0736, weight decay 0.0739.

## 5 Conclusion

This paper presented a framework for evaluating the resilience of Bayesian and Deterministic Convolutional Neural Networks against Single Event Upsets in harsh environments. Contrary to hypotheses regarding the intrinsic resilience of weight uncertainty, our cross-validated evaluations on ShipsNet and EuroSAT demonstrated that deterministic CNNs are strictly more robust to internal parameter corruption than their Bayesian counterparts. Furthermore, our ablation studies revealed a null result for structural regularizers (dropout, smart pooling, weight decay), proving they offer no measurable defense against SEUs. However, the choice of activation function proved critical; bounded and periodic functions (Weighted Gaussian, Sinusoidal) consistently outclassed standard unbounded ReLUs across both datasets.

Our injection site analysis isolated the IEEE-754 exponent MSB (bit 1) as the catastrophic failure point across all models, generating an order of magnitude more damage than mantissa bit flips, with ‘0’ $\rightarrow$ ‘1’ flips being the most destructive. This finding translates directly into a concrete design recommendation for future work: selective radiation hardening. Rather than implementing expensive Error Correcting Codes (ECC) across the entire 32-bit floating-point word, safety-critical systems could selectively harden only the exponent field (or specifically bit 1), achieving massive resilience at roughly 1/32 of the memory and energy overhead. Future work will simulate this selective hardening strategy on larger state-of-the-art architectures.

We scope these claims to the bounds of our experimental design. Our findings rely on 5-fold cross-validation over two satellite imaging datasets and a single CNN architecture family. Additionally, we acknowledge that the Bayesian confidence degradation metric (ASD) is inherently bounded by a Monte-Carlo sampling noise floor. Future research should evaluate deeper networks and explore inference techniques utilizing common random numbers to eliminate baseline sampling variance.

## REFERENCES